Interpretable Multilingual Attribution of Frontier LLMs

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***Abstract*—Most work on telling large language models apart uses English, and the accuracy that interpretable features give up against character n-grams has not been measured on multilingual data with the content held fixed. We collected 840 persuasive essays from five frontier models in seven languages, using 12 prompts with the same content in every language, and described each essay with 21 features computed the same way in every language, using Universal Dependencies parses where syntax is needed. Under leave-one-prompt-out cross-validation, logistic regression on these features identifies the author with 59–72% accuracy in every language (chance is 20%), and most of that accuracy holds for a reading-level instruction the classifier never saw. Character n-grams are 3 to 22 points more accurate within a language, but with the same unlabeled adaptation the features transfer better between languages (49% against 41%). We detect no decline down a coarse resource ranking fixed in advance, though a drop of up to 17 points from English to Hindi cannot be ruled out. Retrained on machine-translated essays, the features still reach 57–70%, while the five models themselves, asked zero-shot, name the author at close to chance (20.3–25.4%). In this setting (one genre and prompt format, one temperature, five known candidates, model-written text), features a person can inspect identify the source model in all seven languages.**

***Index Terms*—authorship attribution, AI-generated text, multilingual text mining, interpretable features, large language models**

1. Introduction

Most work on machine-generated text asks whether a person or a model wrote a text. Provenance also asks which model wrote it, and most of what we know about that comes from English [1], [2]. The two multilingual studies we know of use fine-tuned or probability-based classifiers [3], [4], which give no reason for a decision. A moderator or teacher who already suspects that one of a few known models wrote a Hindi or Turkish text can act on "this text uses paragraphs and punctuation the way model A does" more easily than on an opaque score, and fine-tuned detectors for each language and model version rarely exist outside English.

There are two reasons to doubt that English results carry over. Lower-resource languages have far less available text [5], so models likely see less of them in training and may write less distinctively there, and many stylometric measures are undefined outside English or track the language more than the author. Character n-grams, the most successful single feature type in authorship attribution [6], work in any language but are tied to the script and vocabulary they were trained on.

We study five frontier models in seven languages, hold each essay's content fixed across languages, and describe every text with 21 features computed the same way in every language, the syntactic ones from Universal Dependencies (UD) parses [7]. An analysis plan committed before data collection (Sec. 3.3) asks whether the models can be told apart within each language (**RQ1**), whether attribution weakens as a language's resource level falls (**RQ2**), whether a model's style carries across languages (**RQ3**) and whether the models converge in lower-resource languages (**RQ4**). Whether machine translation removes the signal (**RQ5**) and whether the models can identify the author themselves (**RQ6**) were added before their data were collected, and the n-gram baseline, feature-economy analysis and robustness checks came later.

2. Related Work

Detection and stylometry. Human-versus-machine detectors fine-tuned on English lose about a quarter of their F1 on other languages [8]. For human authors, character n-grams are the most successful single feature [6], masking all but the most frequent words isolates a structural signal [9], UD-derived syntactic features attribute authors in several languages [10], and models tuned to evade human-versus-machine detectors still carry a detectable style [11].

Model attribution. In English, stylometric [12] and embedding classifiers separate neural generators [1], and the latter largely survive LLM translation into Chinese [2]. Multilingual work uses fine-tuned or probability-based detectors: La Cava et al. cover seven older LLMs and human text in 18 languages and find English texts the hardest to attribute [3], and MultiGhostBench covers five recent

LLMs in six languages [4]. Zaitsu et al. attribute Japanese text among six LLMs with part-of-speech and function-word features [13]. To our knowledge, no earlier study of which model wrote a text holds content fixed across languages while using features defined the same way in every language, or compares such features with character n-grams both within and across languages.

Homogenization and LLM judges. Writing with LLMs reduces the diversity of human text [14], but whether models converge on each other as training data thins has not been tested. LLM evaluators favor their own outputs [15], and models asked to name the author of a single English text score near chance and default to GPT or Claude [16]. RQ6 repeats that task in seven languages.

3. Data and Method

3.1. Models, Languages and Prompts

We queried the five subject models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3 and DeepSeek V4 Pro) through OpenRouter in September 2026 and logged the model name OpenRouter returned for every response. The seven languages are English, Spanish, Chinese, Russian, Japanese, Turkish and Hindi. Before data collection we gave them resource ranks 1 to 7 in that order, as a rough proxy for their share of web training data. Joshi et al. [5] put English, Spanish, Chinese and Japanese in their highest class (5) and Russian, Turkish and Hindi one class lower (4) on a 0 to 5 scale, so all seven are well resourced by their measure; we also report that two-class split.

Twelve English prompt schemas each fix a topic, a stance (six for, six against), three supporting claims, a prose-only instruction and a reading level (four prompts each for general, educated and expert readers). For each language, a model that is not a subject (Llama 4 Maverick) wrote a native prompt from the schema rather than translating the English one, so the wording is idiomatic while the content stays the same. A model from a third family (Qwen 3.7 Plus) read each of the 84 prompts blind, extracted the topic, stance and claims, and compared them with the schema. Failed prompts were regenerated until all 84 passed.

3.2. Generation, Validation and Features

Each prompt was sent as a single user turn with no system prompt, at temperature 0.7, twice per cell with fixed seeds and with hidden reasoning excluded, giving $5 \times 12 \times 7 \times 2 = 840$ responses. Each response had to pass language identification (confidence ≥ 0.6), fall within 150 to 900 word equivalents and pass a per-language refusal check. One Grok response to a Hindi prompt was in English and was excluded, leaving 839 texts, and ten DeepSeek responses that used up their token budget on hidden reasoning were collected again with a larger budget.

We parsed the texts with Stanza [17], splitting Chinese and Japanese on sentence-final punctuation first, and computed 21 features in five groups. *Lexical*: moving-average type-token ratio (MATTR, window 50), hapax rate (share of words used once), mean token length, Zipf slope. *Syntactic*: mean and SD of sentence length, burstiness of sentence length, mean maximum dependency depth, subordinate clauses per sentence (`acl`, `advcl`, `ccomp`, `xcomp`, `csubj`), function-word ratio, first-person rate. *Structure*: paragraph count and length, question rate, connective rate. *Punctuation*: commas, colons, dashes and semicolons per 1,000 tokens, with full-width and script-specific variants mapped to the same feature. *Character*: character-bigram entropy and digit rate.

3.3. Analysis

The analysis plan (`docs/analysis_plan.md`) was committed on 3 September 2026 (commit `82673ac`) before any response was collected, and the repository and its history are public. RQ5 was added before any translation existed (`88e08a7`). RQ6 was added before the judge data were collected, after a small pilot that set the reasoning condition (`60593f5`, `c7232ff`).

Validation and tests. The two generations of a prompt are related, and so are its versions in the seven languages. Every split is therefore leave-one-prompt-out (LOPO, 12 folds), every interval is a prompt-clustered bootstrap, the generalized linear models (GLMs) use prompt-clustered errors, and accuracy differences are tested on the 12 per-prompt accuracies with an exact sign-flip permutation test (4,096 sign assignments; the smallest one-sided p is 1/4,096). Holm correction is applied over the 28 headline comparisons (features and n-grams against chance, features against n-grams, and features against the best judge, in seven languages), over the 42 transfer pairs of each representation and over the five judges. Spearman p-values over seven languages are exact.

Classifiers and tuning. Features are standardized inside each training fold (no feature has missing values) and fed to L_2-regularized multinomial logistic regression. The baseline fits TF-IDF over character 1–4-grams within word boundaries (minimum document frequency 2, sublinear term frequency, at most 60,000 terms) on the training prompts only, followed by the same classifier. The features use $C = 1$, as fixed in the plan, and the later n-gram baseline uses the same C with TF-IDF settings that were set once and not tuned. As a check, we also choose $C \in \{0.01, 0.1, 1, 10, 100\}$ for both by nested LOPO on the training prompts.

RQ2. A binomial GLM of cell-level correctness on resource rank, with errors clustered by prompt, by language, and by both. **RQ3.** A per-feature ANOVA (model $\times$ language, with prompt as a block) gives partial η^2. For transfer, features are z-scored within each language (using all texts of that language, without labels) and a classifier trained on language A is tested on language B. The n-gram version fits the vocabulary on the union of A and B without labels and z-scores the TF-IDF columns within each language, so both representations get the same unlabeled adaptation. **RQ4.** After within-language z-scoring,

TABLE 1. RESULTS PER LANGUAGE, IN RESOURCE-RANK ORDER. FEATURES: LOGISTIC REGRESSION ON THE 21 FEATURES, ACCURACY (%) WITH PROMPT-CLUSTERED 95% CI, AND MACRO-F1; n-GRAM: CHARACTER 1–4-GRAM TF-IDF, ACCURACY (%) WITH CI; JUDGES (RQ6): MEAN ACCURACY (%) OF THE FIVE LLM JUDGES, BEST IN PARENTHESES; SEP. (RQ4): MEAN CENTROID DISTANCE IN WITHIN-LANGUAGE SDS. CHANCE IS 20%.

Language	Features [CI]	F1	n-gram [CI]	Judges	Sep.
English	71.7 [61–82]	0.711	75.0 [71–79]	23.0 (26.7)	3.59
Spanish	62.5 [53–70]	0.625	73.3 [65–80]	20.7 (21.7)	3.11
Chinese	65.0 [55–73]	0.644	86.7 [81–93]	20.8 (23.3)	3.01
Russian	65.0 [52–78]	0.654	78.3 [66–88]	23.7 (25.8)	2.97
Japanese	70.8 [63–78]	0.707	79.2 [71–87]	19.8 (22.5)	2.70
Turkish	59.2 [50–69]	0.596	78.3 [73–84]	20.5 (25.8)	2.84
Hindi	63.9 [56–71]	0.634	78.2 [73–83]	24.7 (36.1)	2.69

TABLE 2. SUMMARY OF THE ANSWERS, AND WHEN TO USE WHICH REPRESENTATION (THIS GENRE, PROMPT SET AND MODEL SNAPSHOT).

Question	Answer
RQ1 Within one language?	Yes: 59–72%; 54–76% on unseen reading level.
RQ2 Resource gradient?	No decline found, but a 17-point drop is not ruled out.
RQ3 Across languages?	Partly: 37–59% between language pairs.
RQ4 Convergence?	No by the planned rule; the drop is mostly Grok.
RQ5 Erased by translation?	No: 57–70% on machine translations.
RQ6 Can the models tell?	Near chance (20.3–25.4%); only Claude beats it.
Use case	**Better representation**
Target-language labels	Character n-grams (3 to 22 points better).
Explanation needed	Features (five keep 78–106% of macro-F1).
Cross-language use	Features (49% against 41% for n-grams).
Labeled translations	Character n-grams (63–78% against 57–70%).

we compute the mean pairwise distance between the five model centroids, the silhouette of the model labels and the between/within scatter ratio; the plan calls it convergence only if all three fall with rank and the Hindi interval lies below the English one for each. **RQ5.** Google Translate and an open non-subject model (MiniMax M3) translated all 120 English responses into the other six languages. We measure attribution on the translations, transfer from classifiers trained on the English originals or on native target-language responses, and each feature's Spearman correlation between originals and translations. **RQ6.** Each subject model saw one text at a time, with a shuffled candidate list and no metadata, and was asked which of the five models wrote it (temperature 0, hidden reasoning off where possible), the exact-model task of Bai et al. [16]. **Feature economy.** A nested selection curve (features ranked by mean absolute standardized coefficient on the training fold only, then the top k refit), group ablations, and a character model that sees only punctuation, digits and whitespace [9]. **Fragility.** We train on the eight prompts of two reading levels and test on the four prompts of the third, and we rerun LOPO without the two paragraph features.

4. Results

4.1. Within-Language Attribution

Table 1 shows that the features identify the author in every language, with accuracy between 59% (Turkish) and 72% (English) against 20% chance. Every prompt-clustered interval excludes chance, and in all seven languages the prompt-level test against chance has Holm $p \leq 0.007$. Part of the signal is length. When each feature is residualized on log length within a language, accuracy falls to 44–57%, which is still well above chance.

n-gram baseline. Character n-grams are more accurate, at 73–87%, or 3 to 22 points above the features (13 on average). They lead in 7 to 11 of the 12 prompts in every language except English (4), but with 12 prompts the difference is Holm-significant only in Chinese (Turkish is next, at $p = 0.059$). Choosing C for both by nested LOPO (selected values 0.01–10 for the features, 0.1–100 for n-grams) moves mean accuracy by less than one point for the features and up by 2 points for n-grams, widening the gap to 8–23 points (mean 15), so the default settings do not favor the features. As expected [6], n-grams gain from vocabulary and morphology that the features leave out.

Fragility. Trained on prompts with two of the three reading levels and tested on prompts with the third, the features reach 54–76% (-3 points from LOPO on average) and n-grams reach 66–88% (-5 points). Without the two paragraph features, which a layout instruction would change first, the features reach 51–69% under LOPO and 52–68% across reading levels, and every prompt-clustered interval stays above chance (the lowest bound is 41%).

LLM judges (RQ6). Asked which of the five models wrote a single text, the judges score between 20.3% (DeepSeek) and 25.4% (Claude). Only Claude is above chance (on all 12 prompts; Holm over the five judges $p = 0.001$, next judge $p = 0.096$), and only Claude names itself more often for its own texts than for others' (17% against 7%; 10 of 12 prompts, $p = 0.015$). GPT-5.5 names itself for 82% of texts whoever wrote them. The feature classifier beats the best judge by 28 to 48 points in every language (Holm $p \leq 0.016$). Bai et al. [16] found the same near-chance behavior, biased toward GPT and Claude, in English, and we see it in every language except for Claude in Hindi (36.1%).

4.2. Feature Importance

Averaged over the seven per-language models, the largest standardized coefficients belong to paragraph count, comma rate, character-bigram entropy, paragraph length and Zipf slope. Language explains far more of each feature's variance than the model does (mean partial η^2, or share of variance, of 0.54 against 0.145, with 0.13 for the interaction and 0.16 for the prompt). Model effects are largest for hapax rate (0.38), Zipf slope (0.37), comma rate (0.32), character-bigram entropy (0.31) and paragraph count (0.29) and smallest for subordination, question rate and dependency depth (≤ 0.03), so the models differ mainly in lexical variety, punctuation and paragraphing and very little in clause structure.

A few features do most of the work (Fig. 1a). The five features ranked highest on the training folds reach 78–106% of the full set's macro-F1 (0.50–0.75 against 0.60–0.71). The

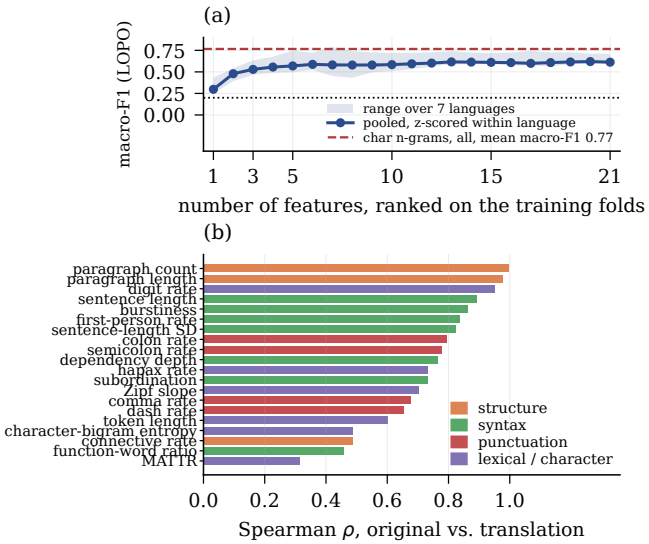


Figure 1. (a) Nested feature-selection curve. Features are ranked on the training folds only, and the band is the range over the seven per-language curves; the dotted line is chance. (b) Spearman correlation of each feature between an English original and its machine translation, averaged over six languages and two translators (lexical and character groups share a color). Question rate is omitted because the English essays contain no questions.

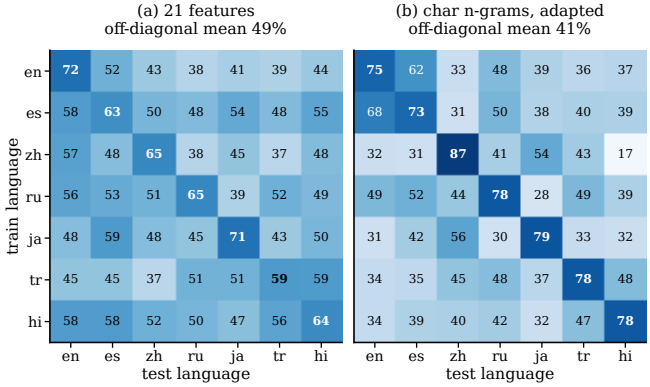


Figure 2. Cross-language transfer accuracy (%). A classifier trained on one language (rows) is tested on another (columns) after within-language z-scoring; the diagonal is LOPO within the language. (a) The 21 features. (b) Character n-grams with the same adaptation (union vocabulary, TF-IDF columns z-scored per language).

ranking depends on the language: paragraph count is first in Russian but sixteenth in English. In the nested selection pooled over languages, the first five picked are paragraph count, paragraph length, sentence length, comma rate and hapax rate. No group of features is essential: removing any one group costs at most 0.11 macro-F1 (punctuation in Hindi, syntax in Turkish). A character model that sees only punctuation, digits and whitespace reaches 49–81% within a language, above the full feature set in Chinese and Hindi, so much of the signal is layout, punctuation and length.

4.3. Transfer and Translation

Cross-language transfer (RQ3). Fig. 2a shows that a classifier trained in one language and tested in another, after z-scoring the features within each language, reaches 37–59% (49% on average, against 65% within a language). All 42 pairs are above chance at the prompt level after Holm correction ($p \leq 0.021$). Character n-grams given the same unlabeled adaptation average 41% (Fig. 2b) and are significantly above chance in 16 of the 42 pairs. The features do better in 33 pairs and on all 12 prompts (paired sign-flip $p = 0.0005$). Since the test language contains the 12 training prompts, these are upper bounds for unseen prompts.

Translation (RQ5). Classifiers trained and tested on the translations (LOPO) reach 57–70%, about 9 points below the English originals and close to native responses in the same languages (64.4% on average), and every prompt-clustered interval excludes chance (lowest bound 48%). Paragraph count ($\rho = 1.00$), paragraph length (0.98), digit rate (0.95) and sentence length (0.89) survive mostly by construction (Fig. 1b), because both translators keep paragraph and sentence boundaries and copy digits, while MATTR (0.32), function-word ratio (0.46) and character-bigram entropy (0.49) are largely set by the target language. A classifier trained on native target-language responses, which has seen none of these texts, scores 52.1% (Google) and 53.1% (LLM) on average. An English-trained n-gram model without vocabulary adaptation scores 20–28% in every language but Spanish (57%), Latin-script Turkish included, so the loss comes from vocabulary.

4.4. Resource Gradient and Convergence

Resource gradient (RQ2). We find no decline along the resource ranking. The cell-level GLM gives $\beta = -0.038$ log-odds per step (SE 0.042, 95% CI -0.12 to $+0.04$; $p = 0.361$ clustered by prompt, $p = 0.226$ by language with seven clusters, $p = 0.343$ by both), less than one accuracy point per step, and the order is not monotone (Hindi is above Spanish and Turkish). Power is limited: the minimum detectable effect at 80% power is about 16 points over the six steps, and the interval allows a decline of up to 17, so we cannot conclude that accuracy is unchanged. Joshi et al.'s classes give the same picture (67.5% for Class 5, 62.7% for Class 4), as does the n-gram baseline (Hindi 78.2%, English 75.0%; GLM $p = 0.380$).

Convergence (RQ4). The models do not meet the convergence rule in the plan. The mean centroid distance falls from 3.59 z-units in English to 2.69 in Hindi ($\rho = -0.96$ against rank, $p = 0.003$; prompt-clustered intervals [3.10, 3.78] and [2.04, 2.86]). The between/within scatter ratio falls from 0.41 to 0.18 ($\rho = -1.00$, $p < 0.001$), but its intervals overlap ([0.19, 0.44] and [0.03, 0.20]), and the silhouette, which also falls with rank ($\rho = -0.93$, $p = 0.003$), stays between -0.02 and 0.04 with overlapping intervals ([-0.05, 0.04] and [-0.10, -0.02]). Only centroid distance separates English from Hindi, and the rule requires all three. Most of the drop from English to Hindi comes from Grok, whose four pairs fall from a mean distance of 4.88 to 2.78, while

the other six pairs average 2.73 in English and 2.63 in Hindi ($\rho = -0.46$ against rank, $p = 0.302$). The decline reflects one model rather than general convergence. La Cava et al. find English the hardest language to attribute [3]; here English separates best.

5. Discussion

Table 2 summarizes when to use which representation. About one prediction in three is wrong, so a label can inform a moderator's judgment but cannot settle it. Neither method can attribute a text to a model outside the candidate set, and we did not test translation combined with a paraphrase attack [11].

Dual use. UD-derived features already attribute human authors [10]; the larger risk is that our rankings show how to hide a model's style. We release model-generated text only.

Limitations. The study covers one genre, persuasive essays with three fixed claims and no lists, so paragraph count and length partly record how each model lays out an argument. All texts were sampled at temperature 0.7 without a system prompt, from model versions served in one month, and we varied the reading-level instruction but not format instructions or temperature; a prompt that dictates the layout could weaken the strongest single cue. Length is entangled with style and is not controlled in the transfer, convergence and translation analyses, and the resource ranking is coarse. Eight features depend on Stanza parses or sentence splitting, whose accuracy varies by language without following the resource order [17], although four of the five top pooled features need no dependency parse. The transfer and translation tests reuse the same 12 prompts, which also limit the power of the prompt-level tests, and no native speaker checked prompt fidelity. All five subjects are closed APIs; we did not test open-weight models, adversarial rewriting or a fine-tuned multilingual encoder [4].

6. Conclusion

In this setting, 21 interpretable features identify which of five frontier LLMs wrote a persuasive essay in seven languages with 59–72% accuracy. Character n-grams are more accurate within a language, but the features transfer better. The signal, led by paragraph count and length, survives translation and holds for an unseen reading level and without the paragraph features. It is a readable check when the candidates are known, the text is model-written and a decision must be explained; we did not compare it with fine-tuned detectors or test it on human text. Next we will test other genres, formats and temperatures, with native-speaker prompt checks and a multilingual encoder baseline.

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AI use: Claude Code (Anthropic) wrote most of the code behind all results, tables and figures, and Claude drafted and edited text in all sections; Sec. 3 lists the LLMs used in the method. The authors reviewed all AI-assisted code and text and take full responsibility for the paper's content. We thank the two anonymous reviewers for their suggestions. Code, prompts, responses, translations, judge calls, features, the analysis plan and result tables: https://github.com/adrian-erlikhman/LangLLM.

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